1. The symplectic two-form $\mathrm{d}\boldsymbol{q} \wedge \mathrm{d}\boldsymbol{p}$.
2. The Hamiltonian (i.e., the energy) H.
3. The volume in phase space (as the vector field is divergence free).
4. The time-reversal symmetry.

A natural question is which of these properties should underpin the design of a numerical method for molecular dynamics? In this section we address this through illustrative examples.

The phase volume conservation can be seen as a consequence of the symplectic property, but it is a weaker condition. It is possible to construct methods that preserve volume but which are not symplectic, and we can build methods that exactly conserve the energy (as we shall show below). Until now we have said little about the time-reversal symmetry of molecular dynamics.

3.6.1 Time-Reversal Symmetry

The term *time-reversal symmetry* or *time-reversibility* is used in two ways in connection with molecular dynamics. In this chapter, we use the term to refer to a property that can be expressed in terms of the individual trajectories of the system. (A somewhat different reversibility property is related to the evolution of probability distributions, see Chap. 8.)

Let $\dot{\boldsymbol{z}} = \boldsymbol{f}(\boldsymbol{z})$ be a dynamical system. By an *involution* we mean a linear mapping $\boldsymbol{z} \mapsto \boldsymbol{R}\boldsymbol{z}$ where $\boldsymbol{R}^2 = \boldsymbol{I}$, i.e. $\boldsymbol{R}$ is its own inverse. Given the involution $\boldsymbol{R}$ we define the *time reversal* of the vector field $\boldsymbol{f}$ with respect to $\boldsymbol{R}$ by

$$\tilde{\boldsymbol{f}}(\boldsymbol{z}) = -\boldsymbol{R}^T \boldsymbol{f}(\boldsymbol{R}\boldsymbol{z}).$$

When a vector field is its own reversal we say that it is a time-reversible vector field.

Let $H(\boldsymbol{q},\boldsymbol{p}) = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$ be the Hamiltonian for a system of $\mathrm{N_c}$ configuration variables. Define the $2\mathrm{N_c} \times 2\mathrm{N_c}$ matrix $\boldsymbol{R}$ by

$$\boldsymbol{R} = \begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & -\boldsymbol{I} \end{bmatrix}.$$

The vector fields involved are

$$\boldsymbol{f} = \begin{bmatrix} \boldsymbol{M}^{-1}\boldsymbol{p} \\ -\nabla U(\boldsymbol{q}) \end{bmatrix}, \quad \tilde{\boldsymbol{f}} = -\boldsymbol{R}\boldsymbol{f}(\boldsymbol{R}\boldsymbol{z}) = -\begin{bmatrix} -\boldsymbol{M}^{-1}\boldsymbol{p} \\ \nabla U(\boldsymbol{q}) \end{bmatrix},$$

which are clearly equal. Therefore the molecular dynamics Hamiltonian system is time-reversible.

The time-reversal symmetry may be understood as a property of the vector field. This engenders a symmetry at the level of the trajectories of the system.

For the system $\mathrm{d}z/\mathrm{d}t = \boldsymbol{f}(z)$, a coordinate transformation $z \mapsto \tilde{z} = \boldsymbol{R}z$ results in

$$\frac{\mathrm{d}\tilde{z}}{\mathrm{d}t} = \boldsymbol{R}\frac{\mathrm{d}z}{\mathrm{d}t} = \boldsymbol{R}\boldsymbol{f}(\boldsymbol{R}^{-1}\tilde{z}) = \boldsymbol{R}\boldsymbol{f}(\boldsymbol{R}\tilde{z}),$$

since $\boldsymbol{R}^{-1} = \boldsymbol{R}$.

A change of time $t \mapsto \tau = -t$ results in

$$\frac{\mathrm{d}z}{\mathrm{d}\tau} = \frac{\mathrm{d}t}{\mathrm{d}\tau}\frac{\mathrm{d}z}{\mathrm{d}t} = -\boldsymbol{f}(z).$$

In other words, for a reversible vector field, the coordinate transformation $z \mapsto \boldsymbol{R}z$ is equivalent to the change of time $t \mapsto -t$.

3.6.2 Time-Reversal Symmetry and Symmetric Numerical Methods

The flow map $\mathscr{F}_t(z_0)$ gives a solution of the differential equation, thus for a time-reversible system, we have, with $z(t) = \mathscr{F}_t(z_0)$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{F}_t(z_0) = \frac{\mathrm{d}z}{\mathrm{d}t} = -\boldsymbol{R}\boldsymbol{f}(\boldsymbol{R}\mathscr{F}_t(z_0)).$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{R}\mathscr{F}_t(z_0) = -\boldsymbol{f}(\boldsymbol{R}\mathscr{F}_t(z_0)),$$

and if we define $\tilde{z}(t) = \boldsymbol{R}\mathscr{F}_t(z_0)$ then $\tilde{z}(t)$ satisfies the differential equation

$$\frac{\mathrm{d}\tilde{z}}{\mathrm{d}t} = -\boldsymbol{f}(\tilde{z}).$$

From this we see that if we start from a point z_0 and integrate forward in time (i.e apply $\mathscr{F}_t$) then apply the involution $\boldsymbol{R}$, we get the exact same result as if we start from $\tilde{z}(0) = \boldsymbol{R}z_0$ and integrate *backward* in time (i.e apply $\mathscr{F}_{-t}$). In sum, since z_0 is an arbitrary point,

$$\mathscr{F}_{-t}(\boldsymbol{R}z) = \boldsymbol{R}\mathscr{F}_t(z).$$

For the flow map, $\mathscr{F}_{-t} = \mathscr{F}_t^{-1}$ so we can recast the identity as

$$\boldsymbol{R} \circ \mathscr{F}_t \circ \boldsymbol{R} \circ \mathscr{F}_t = \mathrm{Id}, \tag{3.14}$$